

Liquid hydrogen refuelling at HRS: Description of sLH2 concept, modelling approach and results of numerical simulations

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ABSTRACT

The paper considers the concept of efficient liquid hydrogen (LH2) refuelling at hydrogen refuelling stations (HRS), presents modelling approach and 3D transient CFD simulation results. The concept is based on the advantages of transforming hydrogen from equilibrium to a non-equilibrium sub-cooled state (sLH2) during compression at pump. The modelling approach comprises a thermodynamic model of LH2 transfer from the HRS tank to the pump exit and a two-phase CFD model from the pump exit through the HRS equipment, i.e. pipes with bends, automatic valve, breakaway, nozzle, and manifold to onboard storage tanks. Due to the absence of published experimental data, the modelling approach and simulations are verified against conceptual LH2 refuelling process available in the literature. The CFD model reproduces key LH2 refuelling parameters: flow rate, pressure, temperature dynamics, including non-uniform temperature in onboard tanks and predicts pipe cooldown from 88K to allowable temperatures corridor of 23.9–26.5 K.

Nomenclature

α	Viscous resistance [N·s/m ³]	S_k	User-defined source terms for k [m ² /s ²]
α_g	Volume fraction of gas phase [–]	S_e	User-defined source terms for ε [m ³ /s ³]
α_l	Volume fraction of liquid phase [–]	t	Time [s]
C_2	Inertial resistance [kg/m ³]	T	Temperature [K]
c_p	Specific heat capacity of the liquid and gas mixture [J/kg/K]	T_g	Gas temperature [K]
E	Energy per unit mass [J/kg]	T_l	Liquid temperature [K]
G_b	Buoyancy terms in turbulence model equations [W/m ³]	T_{sat}	Saturation temperature [K]
G_k	Turbulent kinetic energy generation due to the mean velocity gradient [W/m ³]	u	Velocity [m/s]
G_ω	Generation of ω [W/m ³]	u_i	Velocity component [m/s]
$G_{\omega b}$	Buoyancy terms in turbulence model equations [W/m ³]	u_j	Velocity component [m/s]
h	Enthalpy [kJ/kg]	u_k	Velocity component [m/s]
h_1	Enthalpy of incoming LH2 flow to the pump [kJ/kg]	$\vec{v}$	Velocity vector [K]
h_{2a}	Enthalpy values at the pump exit state in an actual compression process [kJ/kg]	w_a	Actual pump work [J]

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h_{2s}	Enthalpy values at the pump exit state in an isentropic compression process [kJ/kg]	w_s	Isentropic pump work [J]
k	Laminar thermal conductivity [W/m/K]	x_i	Cartesian coordinate [m]
k_{eff}	Effective thermal conductivity [W/m/K]	x_j	Cartesian coordinate [m]
LV	Latent heat of vaporization [kJ/kg]	x_k	Cartesian coordinate [m]
$\dot{m}_{transfer}$	Mass transfer rate between the liquid and gas phase [kg/s]	Y_M	Contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate [–]
$\dot{m}_c$	The mass rate of condensation [kg/s]	Y_k	Dissipation of k due to turbulence [J/s/m ³]
$\dot{m}_e$	Mass rate of evaporation [kg/s]	Y_ω	Dissipation of ω due to turbulence [J/s/m ³]
p	Static pressure [Pa]	Γ_k	Effective diffusivity of k [m ² /s]
Pr_t	Turbulent Prandtl number [–]	Γ_ω	Effective diffusivity of ω [m ² /s]
Q	Energy generated or absorbed due to the phase change [kJ]	ρ	Density [kg/m ³]
r_c	Lee's model coefficients [1/s]	ρ_g	Gravitational force [N]

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r_e	Lee's model coefficients [1/s]	μ	Laminar dynamic viscosity [N's/m ³]
S_k	User-defined source terms [W/m ³]	μ_{eff}	Effective dynamic viscosity [N's/m ³]
S_ω	User-defined source terms [W/m ³]	μ_t	Turbulent dynamic viscosity [N's/m ³]
Acronyms			
CcH2	Cryo-compressed hydrogen	MFR	Mass flow rate
CEP	Clean Energy Partnership	MLI	Multi layer insulation
CFD	Computational fluid dynamics	PCV	Pressure control valve
CGH2	Compressed gaseous hydrogen	PID	Piping and Instrumentation diagram
CV	Control volume	sLH2	Sub-cooled liquid hydrogen
GHG	Greenhouse gas	SoC	State of charge
HVD	Heavy-duty vehicle	TRL	Technology readiness level
HRS	Hydrogen refuelling station	UDF	User defined function
LH2	Liquid hydrogen	VOF	Volume of fluid

1. Introduction

In recent years, hydrogen has gained significant attention as a key green energy carrier [1–3]. The primary reason for the recent surge in interest surrounding hydrogen is the implementation of net-zero programs in different countries, aimed at reducing CO₂ emissions to mitigate climate change and global warming [4,5]. Heavy-duty vehicles (HDVs) will play a significant role in achieving the goals of net-zero greenhouse gas (GHG) emissions, considering the ongoing rise in freight activity [6]. The extent of GHG emissions reduction offered by hydrogen fuel cell HDVs compared to diesel HDVs can be substantial, depending on the feedstock used to produce hydrogen [7].

There are three main options to store hydrogen: compressed gaseous hydrogen (CGH2), cryo-compressed hydrogen (CcH2), and liquid hydrogen (LH2). CGH2 at 70 MPa and 288 K has a density of approximately 40 kg/m³. CcH2 combines compression and cryogenic cooling further increasing its density. LH2 at atmospheric pressure and temperature 20 K has a density of 71.3 kg/m³. Ahluwalia et al. [8] analysed the superiorities of using LH2 storage in vacuum-insulated, low-pressure Type 1 tank for heavy-duty trucks over CGH2 at 35 MPa and 70 MPa in carbon fibre composite Type 4 tanks and they concluded that LH2 storage offers significant benefits, including a 34–54% increase in useable hydrogen storage density, 59–68% higher gravimetric capacity, 25–48% higher volumetric capacity, and 36–41% lower costs. The comparison of refuelling procedures at CGH2 and LH2 HRSs shows that LH2 refuelling is significantly more energy-efficient, with energy consumption of 0.37 kWh/kg for LH2-HRS compared to 2.43 kWh/kg for CGH2-HRS [9].

On the other hand, storing LH2 inevitably leads to its evaporation in time due to heat transfer from outside. The resulting pressure build-up in a closed storage tank requires periodic release of hydrogen from the ullage to exclude failure of the storage tank from overpressure. This phenomenon is referred to as a "boil-off" [10]. The boil-off leads to a 1.5% loss of stored energy per day for small cryostats [11].

Comparing CGH2 and LH2 storage options, Pizzutillo and Acher [12] reported that LH2 not only has more than 50% higher density, i.e. 63 kg/m³ at 1.6 MPa and 28 K against 40 kg/m³ at 70 MPa and 290 K but also requires only an insulated stainless steel low-pressure tank compared to more expensive carbon fibre reinforced high-pressure tanks. The CcH2 storage technologies are at a lower technology readiness level (TRL) that further emphasises the advantages of LH2 storage and fuelling in terms of simplicity and cost-efficiency [13]. Being denser than gaseous hydrogen, LH2 allows for the use of smaller tanks [14]. This results in a lighter storage system which makes it safer for passengers, as high-pressure gas tanks carry the risk of leakage and explosion [15].

Recently, an LH2 refuelling station design was proposed [13], featuring a high flow rate pump of 400 kg/h with a target pressure of 1.6

MPa for the onboard tanks. The aim was to make LH2 refuelling quicker and more efficient. The lower working pressure significantly reduces the cost of ownership for an LH2 HRS compared to a CGH2 HRS while improving the overall efficiency. In 2021, white papers were published [16,17] detailing the fuelling procedure and hardware interface specifications developed through the Clean Energy Partnership (CEP) activities. These specifications are currently undergoing standardization through the ISO/TC197 WG35 "Liquid Hydrogen Land Vehicle Fueling Protocol".

The white paper [17] presents a schematic of an LH2 refuelling station. It includes a vacuum-insulated LH2 tank, an LH2 pump with a maximum mass flow rate (MFR) of 400 kg/h and a maximum pressure of 2.0 MPa, a flow meter, different valves, a breakaway coupling, a hose, and a nozzle. While the design and implementation of each LH2 refuelling station may vary, these components are depicted in Fig. 1.

Maintaining LH2 at temperatures around 20 K restricts options for structural materials suitable for LH2 storage. Aluminium 5083 stands out as the most employed material at cryogenic temperatures, although aluminium 2219 is used as a vessel material for storing cryogenic liquid fuel in launch vehicles [19]. Additionally, austenitic stainless steels like SS304 and SS316 are favoured for cryogenic applications due to their outstanding strength and fracture toughness, which remain robust even at low temperatures [19]. Vacuum jackets incorporating perlite powder or glass bubble insulation systems are deployed in stationary settings, particularly for extremely large LH2 storage tanks [19]. Multi-layer insulation (MLI) stands as the preferred technique for cryogenic steel tanks equipped with vacuum jackets, effectively restricting heat gain to 1–2 W/m². This information can be used as a boundary condition for simulations of vacuum-isolated equipment at LH2 HRS.

To date, the CFD modelling of LH2 tank filling has been focused on the process dynamics in the tank itself. Critical components of the LH2 HRS and their effect on pressure and temperature dynamics were not considered. This hinders the reliable assessments of refuelling process safety and efficiency. In 2022, Wei et al. [20] conducted a numerical simulation and analysis of the LH2 storage tank filling process under sloshing conditions by coupling the sloshing model with the phase-change model. The impact of various sloshing conditions during the filling process by altering the amplitude and frequency of the sloshing was investigated. It was concluded that in sloshing facilitates the LH2 is spreading along the wall, consequently weakening temperature stratification in the storage tank. In 2022, Kang et al. [21] performed a numerical study of the initial charging process of the LH2 tank for vehicles using a 2D CFD model. Only the first 1.2 s of the process were analysed. The results showed that the temperatures of the inner and outer walls and the insulation layer experienced a notably slower temperature decrease compared to the gas region due to their higher heat capacity and insulation effectiveness. In 2023, Ma et al. [22] proposed a four-node mathematical model to examine the no-vent-filling performance of LH2 under microgravity conditions. The results demonstrated that a higher initial wall temperature results in a significant increase of pressure rise in the tank and a noticeable decrease in the final filling level due to early reaching the tank pressure safety limit. Lowering the inlet liquid temperature, particularly at the beginning of fuelling, offers an opportunity to cool the tank and achieve a higher filling level under lower pressures. However, the authors mentioned that the proposed model is less accurate than the full-resolution CFD model for the detailed evolution of physical fields [22]. There are other publications focussing on the boil-off phenomenon [23–25]. Nonetheless, to the best of the authors' knowledge, there are no publications on modelling of the part or entire chain of equipment at LH2 HRS using an efficient refuelling concept called sometimes "sub-cooled liquid hydrogen" (sLH2).

The LH2 pump is a core component of an LH2 refuelling station that can enable the sLH2 fuelling technology if properly implemented. In 2021, Li et al. [25,26] experimented to evaluate the efficacy of a refuelling system installed on a trailer, employing a large-scale,

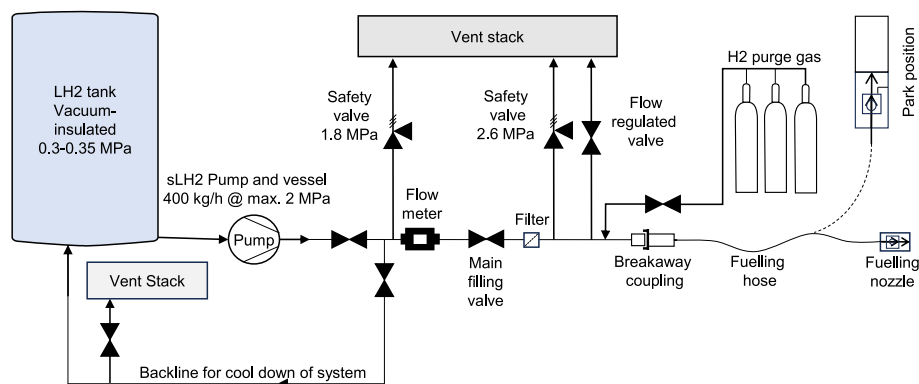


Fig. 1. Schematic of an exemplary LH2 refuelling station [18].

low-frequency, single-acting submerged high-pressure LH2 pump to achieve a flow rate of 285 kg/h. Owing to the challenging cryogenic conditions at about 20 K and pressure fluctuations, precise measurement of internal pump parameters posed significant challenges. Consequently, the study concentrated on the overall refuelling system performance, neglecting investigation into the pump's characteristics. The existing literature lacks knowledge on the modelling of LH2 pumps, impeding deeper insight and optimisation of their functionality [27]. Due to the notable LH2 compressibility, reciprocating LH2 pumps exhibit similarities with reciprocating gas compressors, suggesting that relevant studies on gas compressors can inform LH2 pump analysis [28]. In 2001, Stouffs et al. [29] proposed a model for the thermodynamic examination of reciprocating compressors, emphasising the influence of clearance volume and convective heat transfer on compressor performance. In 2017, Roskosch et al. [30] introduced a thermodynamic differential compressor model tailored for reciprocating compressors, enabling rapid volumetric and isentropic efficiencies prediction. In 2020, Mohammadi-Amin et al. [31] developed quasi-steady 2D CFD and 0D thermodynamic models for reciprocating compressors, exploring performance parameters under near-isothermal conditions. Notably, both models demonstrated similar results, highlighting the quasi-steady model's efficacy in their parametric analysis.

LH2 possesses distinct physical properties compared to gaseous hydrogen [32,33], resulting in the design and operational disparities between LH2 pumps and gas compressors. Firstly, LH2, stored near saturation with low latent heat, necessitates pre-compression in LH2 pumps to prevent cavitation, thus leading to a more complicated pump design. Secondly, during pump operation, hydrogen undergoes compression from near-saturated liquid to high-pressure sLH2, eventually reaching a supercritical state. This complex interplay of temperature, pressure, and phase change fundamentally alters fluid flow and heat transfer within the pump cylinder, diverging from gas compressor behaviour. Interaction between the cylinder wall temperature and hydrogen temperature in LH2 pumps may significantly impact pump performance [34]. Moreover, in-cylinder heat transfer and thermal hysteresis within the pump cylinder wall may induce cavitation due to LH2's proximity to saturation. Consequently, an accurate heat transfer model within the pump cylinder is pivotal for modelling, affecting cylinder temperature and pressure, as well as evaporation loss within the container [28].

During LH2 refuelling, the pump temperature has a huge impact on its performance. In 2024, Kim et al. [27] performed an experimental investigation of a cryogenic piston-type pump operational performance as a function of sub-cooling degree. The results indicated that optimisation of the sub-cooling degree at the design phase can improve the duty cycles of cryogenic liquid pumps. This is the reason that at some LH2 refuelling stations, the pump is submerged in the LH2 stationary storage tank [35].

Sub-cooled at the pump LH2 flows through pipes and valves toward

onboard tanks. Heat transfer and flow losses along the LH2 pathway are unavoidable and occur when the fluid is transferred between vessels [25]. Certainly, the fluid experiences multiple transformations throughout its journey along the entire pathway, including two-phase transitions, sub-cooled liquid phases, super-heated warming, and non-uniform temperature distributions in a two-phase system. Accurate equations of state and proper implementations of two-phase behaviour are crucial to address these complexities [25].

The efficient sLH2 fuelling technology implies filling of vehicle tank with LH2 at a sub-cooled state, i.e. as liquid at a non-equilibrium state above the saturation curve on the P-T phase diagram. This non-equilibrium in the sense of vapour-liquid saturation state can be achieved by an increase of pressure in the pump and a corresponding change of temperature of compressible LH2. In this sub-cooled state LH2 will flow as “non-vaporising” liquid unless heat transfer from warmer equipment, specifically at the start of refuelling, and from outside through the components of the refuelling system will “return” LH2 from sub-cooled conditions to the saturation curve conditions (afterwards a notable LH2 evaporation is expected because of further heat transfer to the two-phase hydrogen system from the outside). Sub-cooled state could be achieved also when LH2 temperature is below its saturation temperature for the considered pressure.

The conceptual mass flow rate, and typical pressure and temperature profiles during efficient LH2 refuelling (sLH2 refuelling) are shown in Fig. 2 [12]. The procedure stems from experimental observations and is subdivided into steps. The pre-fuelling activities include purging and leakage testing (not shown in the diagram), and system depressurisation. During the depressurisation step, the temperature drops due to vapour phase expansion and liquid phase cooling due to evaporation accompanying the pressure reduction. The main refuelling consists of three steps: one with a reduced mass flow rate (10–20% of the maximum flow rate) to cool down piping and storage systems, then refuelling with a targeted average fuelling rate of 400 kg/h, and finally the end of fuelling with staged reduction of mass flow rate to zero when pressure reaches 1.6 ± 0.05 MPa. The post-fuelling involves additional purging and leakage testing after reaching the target pressure, before disconnecting the nozzle.

The conceptual temperature corridor after cooling down is $23.9 \text{ K} \leq T \leq 26.5 \text{ K}$. The draft of the fuelling protocol which is under development by ISO/TC197/WG35 expects the supply of sLH2 to ensure the cooling down of warm components along the supply line and the condensation of gaseous hydrogen in the LH2 HRS. This would allow us to avoid prematurely reaching the target pressure limit. The HRS should target an average temperature of the supplied LH2 below 26.5 K. The reference point for this temperature is in the dispenser within about 4 m from the refuelling assembly. For safety reasons, the HRS should consider means of monitoring the temperature and define appropriate actions in case of deviation from this value.

One of the important advantages of the sLH2 refuelling technology is

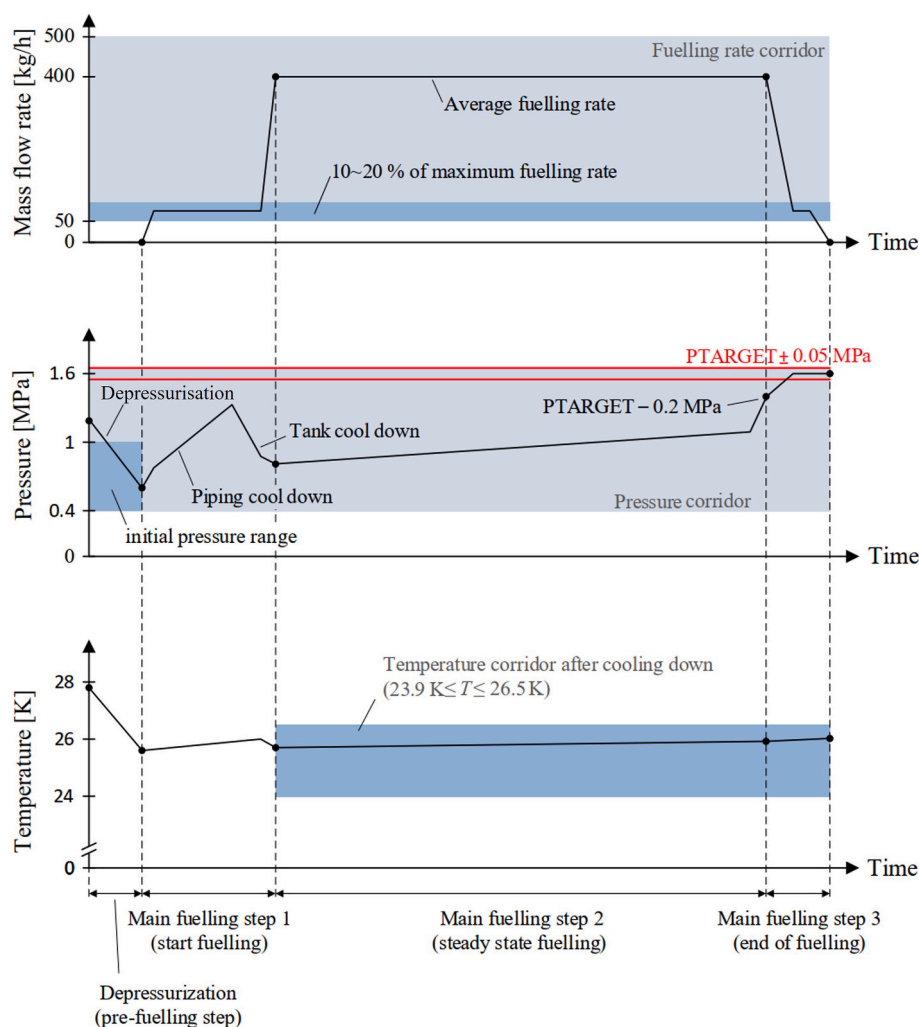


Fig. 2. Conceptual mass flow rate, pressure, and temperature patterns during LH2 refuelling [13].

that only one nozzle is required as there is no gaseous hydrogen returning from onboard storage. This essentially simplifies the handling of LH2 refuelling [12]. The technology also reduces the complexity of the refuelling hardware and the fuelling protocol, compared to CGH2 and CcH2 storage systems. This makes it the technology of choice for efficient hydrogen storage [5–8]. The absence of contemporary models for LH2 HRS hinders the understanding of underlying physical phenomena and creation of inherently safer refuelling protocols capable of accounting for details of the station's equipment, hydrogen storage parameters, and environmental conditions.

This paper addresses the critical need for efficient LH2 refuelling protocols and standards, in light of the growing global demand for hydrogen as a clean energy carrier. Currently, most of hydrogen refuelling research deals with design and optimisation of CGH2 HRS, and while some research has explored LH2 storage, there remains a significant gap in the modelling and validation of the LH2 refuelling process across the entire refuelling station equipment. This study fills this knowledge gap by developing a comprehensive modelling approach that captures the thermodynamic and fluid dynamics aspects of the LH2 refuelling process, from the HRS storage tank through the HRS equipment to the onboard storage system. The study integrates the thermodynamic modelling with 3D two-phase CFD simulations to reproduce the complex refuelling process. Unlike existing models that focus primarily on gaseous hydrogen [36–38], this paper presents for the first time full-scale simulations of the LH2 refuelling process.

2. Problem formulation and modelling approach

The design of the conceptual LH2 fuelling station is presented in Fig. 1 [17]. The modelling of LH2 refuelling parameters from the HRS tank to the pump is conducted in this study by using the thermodynamic model (under the next paragraph assumptions), and simulations of the process from the pump to onboard storage tanks are carried out by contemporary 3D two-phase CFD model.

The following assumptions are adopted for the modelling in the absence of publicly available LH2 refuelling experiments. The HRS storage tank has a volume of 61.36 m³ and can hold 4 tonnes of LH2 at pressure 0.3 MPa ab and a saturated temperature of 24.68 K. Non-uniformity of temperature in stationary HRS tank is a separate challenging problem that is not considered here, and temperature of hydrogen liquid and gas is assumed to be equal. Moreover, for the purpose of this study, it is assumed that the temperature within the larger HRS tank remains constant throughout the LH2 transfer process. This assumption is justified by the substantial difference in volume between the HRS tank and onboard tanks. The larger tank's considerable thermal mass ensures that any heat transfer or temperature variation within it is negligible compared to the smaller onboard tanks. The hydrostatic pressure in the HRS tank is neglected in the thermodynamic model of the LH2 transfer process from the HRS tank to the pump exit as a trivial issue.

There are two onboard storage tanks, each is of 0.71 m³ and can accommodate 44 kg of LH2 at 100% state of charge (SoC). The volume of

the onboard tank was calculated based on the designed capacity of 44 kg in the assumption of LH2 density equal to 63 kg/m^3 [17] at a final pressure of 1.6 MPa and temperature of 29 K. The tank shape is a simple cylinder with two hemispheres on either side of the tank (Fig. 3). Based on the assumed length-to-diameter ratio $L/D = 3.5$, the internal tank length is estimated as 2.3 m and the internal diameter as 0.66 m. The tanks are assumed to be installed on HDV vertically and filled from the top. Tank internal walls are made of stainless steel of 3 mm thickness. Onboard tanks are initially at 0.6 MPa ab pressure. An equal temperature of 28.25 K (saturated temperature for 0.6 MPa ab pressure) is assumed for both LH2 and hydrogen gas in the ullage. The initial LH2 filling level in onboard tanks is 35% by volume, i.e. 75% of ullage space.

The tanks are assumed to be insulated using an MLI. The heat flux through the insulation system, including the MLI material and vacuum jacket, depends on the tank design. The value of heat flux of 1 W/m^2 is applied to all tank walls based on the review of published data, e.g. by Zheng et al. [39]. The experimental cumulative heat transfer rate at the breakaway coupling to the receptacle is 15 W [18]. Assuming that this is heat flux from the coupling to hydrogen flow and considering the 25 mm diameter pipes and the 0.1 m length of the breakaway, the heat flux can be calculated using the surface area formula ($A = \pi DL$) as follows: $q = Q/\pi DL$. This yields a heat flux of approximately 636.6 W/m^2 . However, in the real world, this value might be much smaller, as the breakaway length might be more than assumed here 0.1 m.

The vacuum-insulated hose is assumed to be 2 m long, and the nozzle length is taken as 30 cm. The sizing of the pipes is depicted in Fig. 4. All pipes, breakaway, hose, and nozzle are assumed to have a 25.4 mm inner diameter made of stainless steel 316L and 2 mm wall thickness (except nozzle with wall thickness of 37.3 mm). The heat flux to the surface of vacuum-insulated pipes was assumed at the lowest imagined level of 0.6 W/m^2 in these particular simulations and applied for all pipes, hose and nozzle. Simulations with higher values of heat flux per unit area need to be carried out as a part of follow-up parametric studies.

The LH2 storage reservoir at HRS and all components down to the pump are assumed to be filled in by LH2 at 0.3 MPa ab pressure and saturation temperature of 24.68 K. The components between the pump and the onboard tanks, i.e. nozzle, breakaway, hose, and connection pipes, are assumed to be filled with gaseous hydrogen. The initial pressure in piping downstream the pump is assumed the same as in

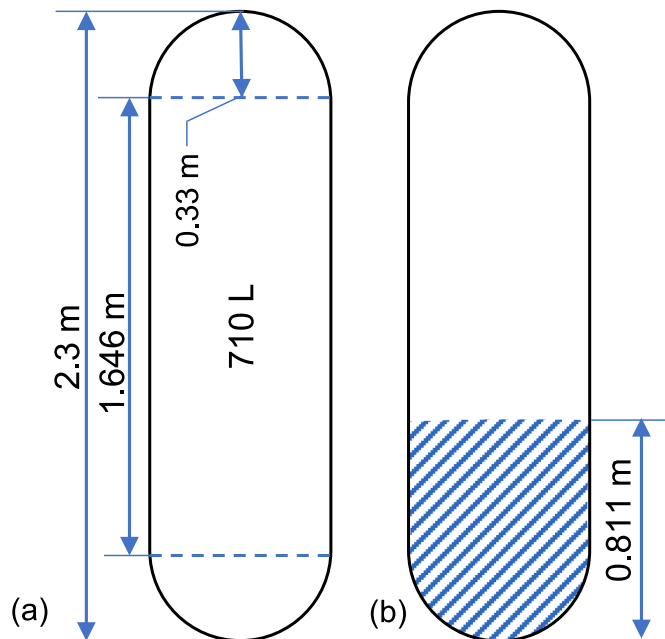


Fig. 3. One of two onboard tanks: (a) dimensions, (b) assumed initial LH2 volumetric level of 35%.

onboard storage tanks, i.e. 0.6 MPa ab. However, a higher initial temperature of piping of 88 K is assumed (higher than the saturated temperature of 28.25 K for this pressure and even higher than the critical point temperature of 33.2 K above which LH2 cannot exist). This is to reflect that the pipes were pre-heated by external heat flux before the start of refuelling. LH2 is assumed to be compressed with 65% isentropic efficiency to 2 MPa while reaching a temperature of 27.01 K. The calculation of temperature in the pump will be described in the next section. The dimensions, initial and boundary conditions for the entire system before fuelling are summarised in Table 1.

Because of the complexity of the inclusion of the pump into the CFD model, the behaviour of hydrogen in the HRS reservoir will not be simulated as well. For the purposes of the thermodynamic modelling, the pump inlet temperature is assumed to be equal to the HRS tank's initial condition during the entire refuelling procedure.

2.1. Thermodynamic model of refuelling process from HRS tank to the pump exit

The LH2 from the HRS reservoir comes to the pump to be compressed to the sLH2 state. This is possible as compression time is comparatively short, and the process is often assumed as adiabatic, i.e. without heat exchange with the environment (especially for the pump submerged into LH2). Though the process in a real pump is polytropic, in the present study the compressed LH2 temperature will be estimated based on the pump isentropic efficiency definition. The isentropic process is the ideal reversible adiabatic process with no heat transfer between the pump or environment. The isentropic efficiency of a pump is defined as the ratio of the work required to increase the fluid pressure up to a specified value in an isentropic process to the actual required work input:

$$\eta_c = \frac{w_s}{w_a} \quad (1)$$

Assuming negligible change in kinetic and potential energies, the work input for an adiabatic process would be equal to the change in enthalpy [40]:

$$\eta_c \cong \frac{h_{2s} - h_1}{h_{2a} - h_1} \quad (2)$$

where h_{2a} and h_{2s} are the enthalpy values at the pump exit state in an actual and isentropic compression process respectively, and h_1 is the enthalpy of incoming to the pump LH2 flow.

Based on the conceptual refuelling protocol [12], the hydrogen temperature at the dispenser (a few meters from the onboard tank inlet) should be in the corridor of 23.9–26.5 K (see Fig. 2). At the station, the pump usually works at its optimum design pressure and the pressure ramp or mass flow rate is controlled via an automatic pressure control valve (PCV). Estimating the temperature at the pump exit is essential for reliable CFD simulations downstream of the pump through the HRS equipment down to the onboard storage tanks.

The efficiency of LH2 pumps is in the range of 55–80% [41]. Using the NIST real gas model [42] and assuming the pump's efficiency of 65%, the LH2 temperature at the pump exit may be estimated by Eq. (2). It is assumed that the LH2 in the HRS reservoir, located upstream of the pump, is at pressure 0.3 MPa ab and saturation temperature of 24.68 K. The maximum pump pressure is 2.0 MPa [16,17]. Therefore, the temperature at the pump exit is calculated as 26.98 K. The expansion of pump-compressed LH2 in the downstream pipes, which are initially at 0.6 MPa ab, is assumed to be adiabatic due to the rapid nature of the process and the pipe insulation. The calculations show that the LH2 temperature after the expansion is 25.98 K, i.e. within the recommended hydrogen temperature corridor of 23.9–26.5 K.

To examine the effect of pump efficiency on the LH2 temperature after the pump, calculations were performed using various efficiencies. The LH2 is compressed from a saturated condition in the HRS reservoir

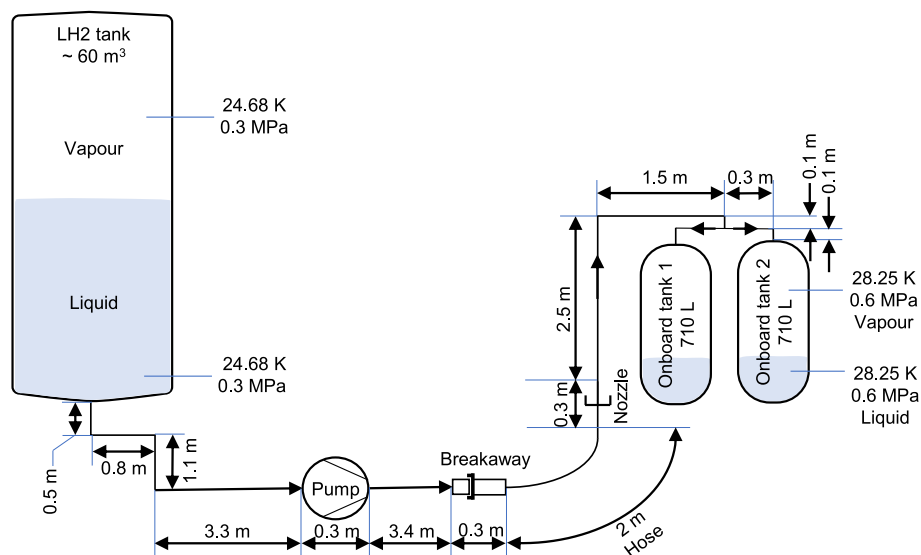


Fig. 4. The PID diagram of the conceptual LH2 refuelling station, including the initial tanks condition, used for modelling and simulations in this study.

Table 1

The dimensions of HRS equipment, and their initial (pressure, temperature, phase fraction) and boundary (heat flux) conditions.

	HRS Reservoir	Pipe 1	Pump	Pipe 2	Breakaway	Hose	Nozzle	Pipes 3, 4	Tanks 1, 2
Length [m]	10	0.5/0.8/1.1/3.3	0.3	3.4	0.3	2	0.3	2.5/1.5/0.1/0.3/0.1	2.3
Diameter [m]	2.78	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.66
P_i [MPa (g)]	0.3	0.3	2.0	0.6	0.6	0.6	0.6	0.6	0.6
T_i [K]	24.68	24.68	27.01	88.0	88.0	88.0	88.0	88.0	28.25
Heat flux [W/m ²]	1.0	0.6	N/A	0.6	636.6	0.6	0.6	0.6	1.0
LH2 fraction ^a	N/A	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.35

Note.

^a Volumetric fraction of liquid phase in equipment.

to the maximum pump pressure of 2.0 MPa ab and then expands to the pipe space that is at a pressure of 0.6 MPa ab. Fig. 5 presents the comparison of the hydrogen temperature change for pump efficiencies of 35%, 65%, 80% and 100%. As the pump efficiency decreases, the temperature of hydrogen after expansion in the downstream pipe space increases. For pumps with efficiencies 65%, 80% and 100%, the LH2 in the pipe downstream of the pump is at a sub-cooled state (sLH2), i.e. above the saturated state curve in Fig. 5. However, for pump efficiency of 35%, the temperature of LH2 after expansion in the downstream pipe reaches the saturated temperature of 28.25 K for 0.6 MPa ab, which is

above the recommended hydrogen temperature corridor (see Fig. 2). LH2 at these conditions is not anymore sub-cooled and, thus, unacceptable for the implementation of the sLH2 refuelling technology due to the “uncontrolled” generation of gaseous hydrogen and thus increase of pressure (with a decrease in refuelling efficiency and potentially serious safety implications). Therefore, pump efficiency is an important parameter when designing LH2 refuelling station equipment and protocols.

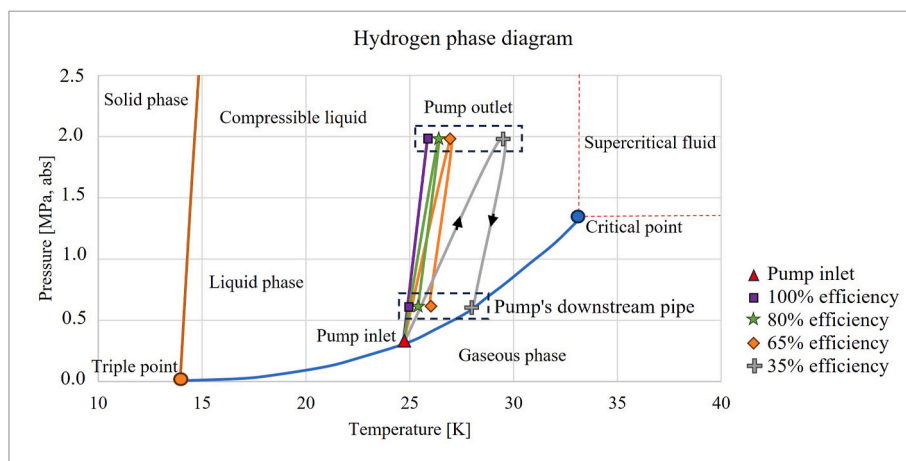


Fig. 5. The phase diagram of hydrogen and results of the thermodynamic model calculations of hydrogen parameters at the inlet to the pump from the HRS reservoir, pump outlet and in space of piping downstream of the pump.

3. CFD model and numerical details

3.1. Geometry and calculation domain

The top view and side view of the HRS geometry from the inlet to the refuelling equipment from the HRS tank to the onboard storage tanks are shown in Fig. 6. The calculation domain for CFD simulations was from the pump exit through equipment to the onboard storage tanks.

Tetrahedral control volumes (CVs) are used to discretise the onboard tanks and the three-way manifold while hexahedral CVs are used in the pipes. The transition from tetrahedral to hexahedral mesh is performed using pyramidal CVs. The entire computational domain is meshed using 239,111 CVs with a minimum orthogonal quality of 0.5 and an average of 0.85. The numerical mesh is generated using ANSYS Mesher. The minimum and maximum CV sizes in the onboard tank are 0.5 cm and 5 cm respectively while the CV size growth rate is maintained at 1.2. Each tank contains 60,523 CVs. Fig. 7 shows the mesh in the onboard tanks and the three-way manifold.

The grid independence analysis is performed using three different meshes: coarse mesh with 104k control volumes (CVs), medium - 239k, and fine - 452k. Each mesh was evaluated to determine its impact on the pressure and temperature dynamics within onboard storage tanks. The analysis aims to ensure that the chosen mesh resolution accurately captures the critical thermodynamic behaviour and provides reliable results for the modelling of LH2 refuelling processes. Fig. 8 (left) displays the pressure in Tank 1 for the three different mesh resolutions. The results for the medium and fine meshes are identical, while the coarser mesh shows a deviation from the start of simulations. A similar result is obtained for the temperature in Tank 1 (Fig. 8, right). The mesh sensitivity analysis in Tank 2 (not shown) demonstrates the same results. Therefore, the medium mesh with 239k CVs is selected for the simulations.

3.2. Governing equations and numerical details

The volume of fluid (VOF) model is applied to simulate the two-phase hydrogen liquid-gas flow [43]. This selection is based on its ability to accurately represent gas-liquid interfaces in the flowing fluids [44–46] yet being computationally affordable. The model assumes that

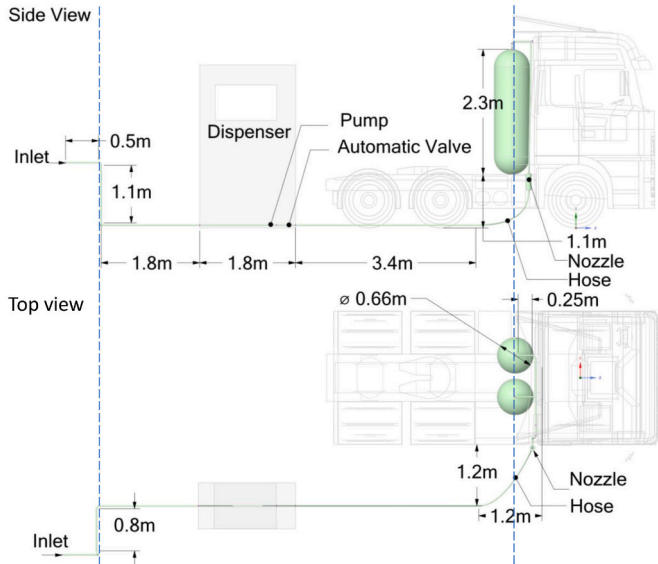


Fig. 6. The side and top views of the LH2 fuelling station geometry, including hydrogen-filled components highlighted in green colour. The calculation domain does not include equipment before the pump exit. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the phases share the same velocity and temperature. Hence, there is no need to solve momentum and energy transport equations for each phase. Instead, the equations are solved for the entire fluid with different phases. The volume fraction of each phase is tracked and monitored across the entire calculation domain [47,48].

The fraction of liquid phase (subscript “l”) is found from the solution of the liquid phase continuity equation written as 48:

$$\frac{\partial}{\partial t}(\alpha_l \rho_l) + \nabla \cdot (\alpha_l \rho_l \vec{v}_l) = \dot{m}_c - \dot{m}_e, \quad (3)$$

while the fraction of the gas phase (subscript “g”) is found based on the condition in the two-phase fluid. The sum of liquid and gaseous volume fractions is always the unity:

$$\alpha_g + \alpha_l = 1. \quad (4)$$

A single momentum equation is solved:

$$\frac{\partial}{\partial t}(\rho_m \vec{v}) + \nabla \cdot (\rho_m \vec{v} \vec{v}) = -\nabla p + \nabla \cdot [\mu_{eff} (\nabla \vec{v} + \nabla \vec{v}^T)] + \rho \vec{g}, \quad (5)$$

where $\mu_{eff} = \mu + \mu_t$.

The energy equation is also shared among the liquid and gas phases:

$$\frac{\partial}{\partial t}(\rho_m E) + \nabla \cdot (\vec{v} (\rho_m E + p)) = \nabla \cdot (k_{eff} \nabla T) + Q, \quad (6)$$

where $k_{eff} = k + \frac{c_p \mu_t}{Pr_t}$, and Pr_t is equal to 0.85 in the current model. The energy per unit mass E :

$$E = \frac{\alpha_g \rho_g E_g + \alpha_l \rho_l E_l}{\alpha_g \rho_g + \alpha_l \rho_l}, \quad (7)$$

in which the specific energy for each phase is:

$$E_j = h_j - \frac{p}{\rho_j} + \frac{v^2}{2}, \quad (8)$$

where h_j is the enthalpy of a j -th phase and j stands for j -th phase, i.e. gas or liquid.

Q is calculated for every CV at each time step, and Q would be a source or sink term depending on the dominant phase change which could be either condensation or evaporation:

$$Q = -\dot{m}_{transfer} \cdot LV, \quad (9)$$

where $\dot{m}_{transfer} = (\dot{m}_e - \dot{m}_c)$. The Q term would be negative in case of evaporation as energy is required to evaporate liquid and positive in case of condensation. The latent heat LV is the difference between the enthalpy of gas and liquid phase at saturation conditions and can be determined as follows [49]:

$$LV = h_g - h_l. \quad (10)$$

In this study, while assuming that hydrogen is in its equilibrium state, LV values are taken from the NIST database [42].

3.3. Turbulence model

ANSYS Fluent [50] recommends using the SST k-omega turbulence model for VOF multi-fluid applications. The reason is that the k-omega model supports turbulence damping in interfacial cells, which is vital for capturing interfacial instabilities [50]. This model was able to achieve a reliable prediction of pressure build-up in tanks for cryogenic and multiphase flows [24,51,52]. The turbulent kinetic energy and its dissipation rate can be obtained from the equations:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_k \frac{\partial k}{\partial x_j} \right) + G_k - Y_k + G_b + S_k, \quad (11)$$

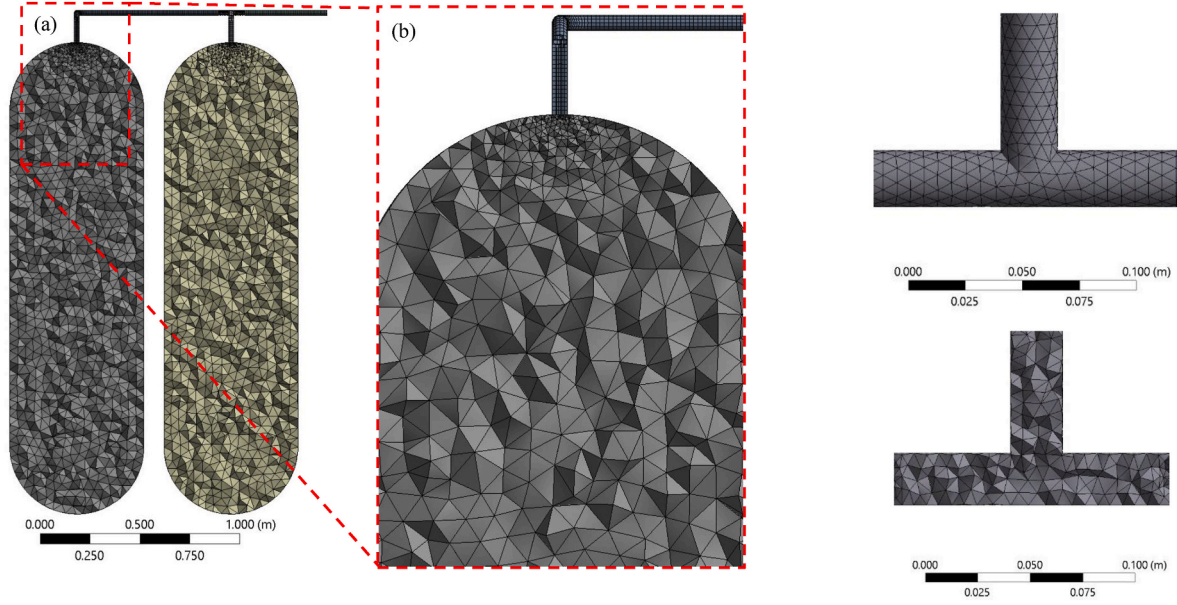


Fig. 7. (a) Central cross-section of the numerical mesh for the onboard tanks; (b) enlargement of the numerical mesh cross-section at the onboard tank inflow; (c) the surface mesh of the three-way manifold; (d) central cross-section of the numerical mesh in the three-way manifold.

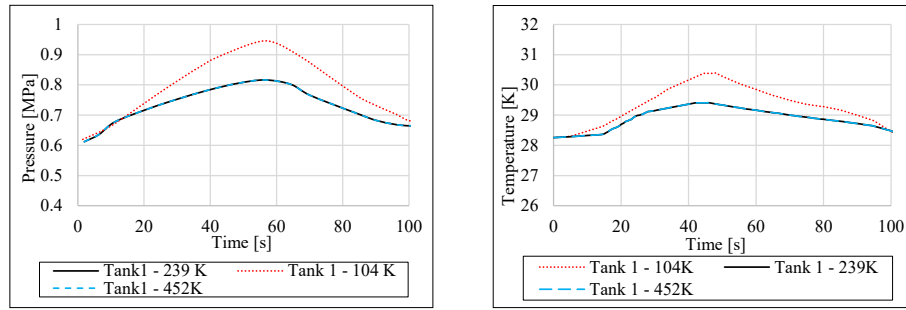


Fig. 8. Impact of mesh resolution (coarse, medium, fine) on pressure (left) and temperature (right) dynamics in Tank 1 of the LH2 storage system during refuelling.

$$\frac{\partial(\rho\omega)}{\partial t} + \frac{\partial}{\partial x_i}(\rho\omega u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_{\omega} \frac{\partial \omega}{\partial x_j} \right) + G_{\omega} - Y_{\omega} + G_{\omega b} + S_{\omega}, \quad (12)$$

where, u_i is i -th component of velocity in cartesian coordinates, and x_i is i -th cartesian coordinate axis [53].

To keep the efficiency and convergence of the calculation, the standard wall functions [54] are utilised for the near-wall region modelling.

3.4. Phase change model

To model evaporation and condensation, Lee's model is used in this study [55]. For condensation, when the saturation temperature is higher than the gas temperature ($T_{sat} > T_g$) the mass transfer rate from gas to liquid [kg/m³/s] is:

$$\dot{m}_c = r_c \alpha_g \rho_g \frac{T_{sat} - T_g}{T_{sat}}, \quad (13)$$

and for evaporation ($T_l > T_{sat}$) the mass transfer rate from liquid to gas is:

$$\dot{m}_e = r_e \alpha_l \rho_l \frac{T_l - T_{sat}}{T_{sat}}. \quad (14)$$

Kharangate and Mudawar [56] stated that Lee's model [55] is developed to streamline the numerical simulations of evaporation and

condensation phenomena while aiming to simplify the computational aspects involved in these processes and the model is not intended to be used for detailed descriptions of phase change mechanism. Lee's model coefficients vary with the thermal properties of the fluid, saturation temperature, mesh grid size and even set-up of the CFD model [56]. Therefore, evaluation and calibration are required to use these coefficients depending on specific problems and conditions [57]. In the current study, Lee's coefficient of 0.05 is selected, following a work by Kangwanpongpan et al. [57] which demonstrated that a value of 0.05 can provide a good agreement of the pressure recovery phenomenon during a large-scale cryogenic tank flash-boiling.

3.5. Initial and boundary conditions

Initial conditions for LH2 volume fraction, temperature and pressure in the pipes and onboard tanks correspond to those presented in Table 1. The temperature in the onboard tanks is assumed to be equal to the saturation temperature at 0.6 MPa ab, i.e. $T_{sat} = 28.25$ K. The LH2 volume fraction in the pipes downstream of the pump is zero as the pipes are initially warmed up to 88 K which is above the critical temperature of 33.2 K for liquid phase existence. The volume fraction of LH2 in onboard tanks is 1 up to the height of 0.526 m from the bottom (35% tank fill) and is 0 above that level in the ullage. Pressure after the pump and down to the onboard tanks is 0.6 MPa ab. The heat fluxes at the pipe walls were used as thermal boundary conditions and presented in Table 1. The values are typical for the LH2 tanks with multi-layer

insulation blankets and vacuumed space between inner and outer walls, e.g. 1–2 W/m² reported by Ahluwalia et al. [19].

3.6. Numerical details

The simulations were performed using ANSYS Fluent 2023R1 as a CFD engine [50]. The pressure-based implicit solver is used to solve the governing equations. The SIMPLEC algorithm was applied for pressure-velocity coupling. Convective terms and volume fractions were discretised using a second-order upwind numerical scheme. The PRESTO discretisation scheme is utilised for pressure discretisation.

The explicit formulation for volume fraction in the multi-phase model is used while having a “sharp interface” modelling option between the liquid and vapour phases. The time step was initially set as 0.001 s, and then gradually increased to 0.01 s to maintain convergence of the simulations. The mass imbalance over the entire fuelling period is less than 1%, which is considered an indicator of good convergence and hydrogen conservation in the calculation domain. The simulation of 525 s of LH2 refuelling time takes about 14 days on a 72-core CPU workstation.

3.7. Modelling of pressure control valve (PCV)

The automatic PCV is represented in the CFD model as a porous media model and the mass flow rate (MFR) as per the conceptual fuelling protocol is achieved in simulations by varying the media porosity. The porous media model integrates an empirically determined flow resistance within a designated ‘porous’ region of the model. Essentially, this model introduces a momentum sink into the governing momentum equations [53]. The momentum sink term includes two parts: a viscous (Darcy) loss term, i.e. the first term on the right-hand side of Eq. (15), and an inertial loss term, i.e. the second term on the right-hand side of the equation:

$$S_i = - \left(\frac{\mu}{\alpha} u_i + C_2 \frac{1}{2} \rho |u| u_i \right). \quad (15)$$

The coefficients α and C_2 in this sink term provide the control of pressure drop at the valve and thus allows controlling MFR during simulations based on its prescribed value. The coefficients α and C_2 were introduced in simulations using the User Defined Function (UDF) capability of ANSYS Fluent. The UDF is programmed to compute simulated MFR through the automatic valve and compare its value with the prescribed MFR to manipulate the porous media coefficients. The heat transfer in the automatic valve is disabled, allowing the porous model to function at equal enthalpy before and after PCV, like the behaviour of a real throttling valve [58].

4. Results and discussion

In the absence of publicly available experimental data for the CFD model validation, the numerical simulations were verified against the conceptual refuelling procedure [12]. The conceptual MFR (Fig. 2, top) is digitised and is used as an input to the UDF to regulate the porosity in the PCV to match the MFR in the conceptual refuelling protocol. Fig. 9 compares the conceptual MFR versus the simulated MFR. It can be realised that the UDF [58] successfully reproduced the conceptual MFR during the whole refuelling process with less than 0.1% deviation. This UDF can reproduce the arbitrary MFR profile and thus can be used in future for the design of HRS equipment and efficient and inherently safer LH2 refuelling protocols.

The temperature after the PCV should be in the temperature corridor 23.9–26.5 K [17]. In the simulations, the maximum working pressure of the pump was set to 2.0 MPa. Under these conditions, the temperature of sLH2 after expansion from the pump to the pipe space is calculated by the thermodynamic model (Eq. (2)) as 25.98 K. This temperature was

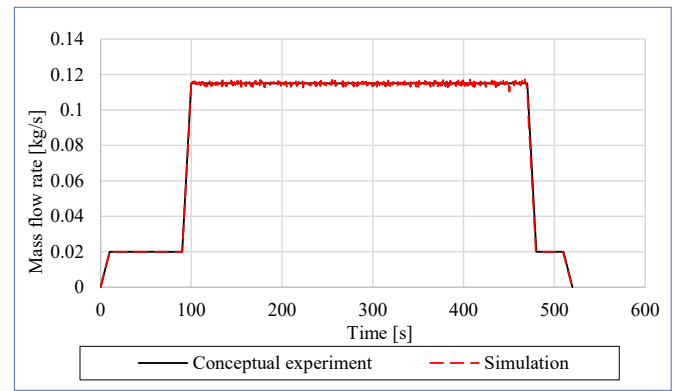


Fig. 9. The conceptual MFR versus simulated MFR.

used as the inlet temperature for the CFD model presuming that the pump operates at its maximum working pressure. The temperature after the PCV is calculated dynamically, based on assumptions of no heat transfer at the PCV, and isenthalpic expansion process in the PCV. The simulated temperature after the PCV is shown in Fig. 10 (solid black line). During the first stage of the main refuelling process, where the MFR is limited to 20 g/s, the pipes, initially at 88 K, begin to cool down to the temperature of sLH2 of 25.9 K within the first 100 s (see Fig. 10). The temperature at different sections of the refuelling line decreases from an initial 88 K to the acceptable range of 23.9–26.5 K (highlighted by light green colour) during the main refuelling process. Then, throughout the entire refuelling process, the temperature remains within the specified limits and practically does not change. At the end of the refuelling process, when the pressure reaches 1.6 MPa, the temperature after the PCV rises to 26.15 K, which remains within the allowable temperature range. However, this increase in temperature might not be evident in Fig. 10 due to the scale of the figure.

It is worth mentioning that the temperature in the refuelling line measurement points is calculated using a mass-weighted average method over volume to account for the temperatures of both the liquid and vapour phases in regions not fully occupied by either phase.

Figs. 11 and 12 show the pressure and total temperature dynamics in the onboard storage LH2 tanks. The total (or stagnation) temperature accounts for the kinetic energy of the fluid being converted to thermal energy as it is brought to rest isentropically. However, at low velocities, the difference between static and total temperature is negligible. Therefore, for these analyses, the temperature refers to the total temperature.

The initial pressure in both onboard storage tanks is 0.6 MPa ab. Fig. 13 displays screenshots of the temperature and vapour volume fraction contour plots in the cross-section of the onboard tanks at various

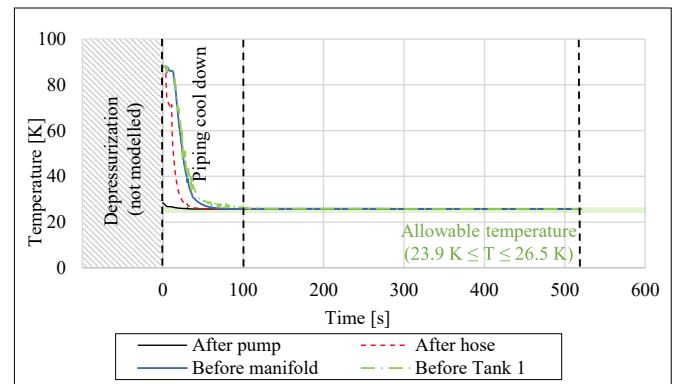


Fig. 10. The simulated mass-averaged hydrogen temperature at different locations of the refuelling line downstream of the pump in time.

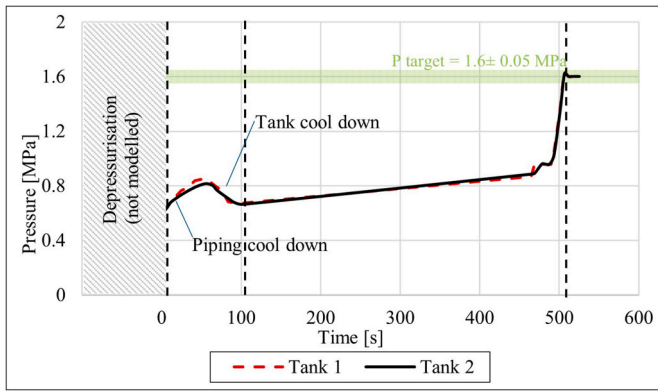


Fig. 11. The simulated pressure in the onboard storage tanks.

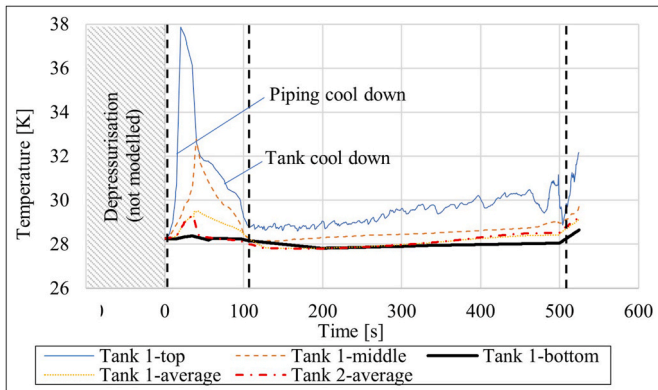


Fig. 12. The simulated temperature in the onboard storage tanks.

stages of refuelling: (a) the initial time (0 s), (b) during step 1 of the main refuelling when the pipes are still warm and LH2 evaporates in the pipes (15 s), (c) in the middle of fuelling step 2 when pipes are cooled down and filled entirely by sLH2 (350 s), and (d) at the final stage of refuelling when the left tank (Tank 1) is filled entirely and the right tank (Tank 2) is near its 100% fill (520 s). The temperature non-uniformity at the end of the refuelling process reaches 3 K for both tanks. It is worth mentioning that Tank 1 is filled earlier than Tank 2, as Tank 1 is closer to the manifold and hydrogen flow reaches this tank first, hence it is filled earlier.

During step 1 of the main refuelling, the pressure inside the tanks increases as the pipes and other refuelling line components are initially at temperatures of 88 K, i.e. substantially higher than the saturation temperature for this pressure, see Fig. 13(a). The sLH2 flows at this stage through these pipes at a limited rate of 20 g/s, causing LH2 evaporation within the pipes before entering the onboard tanks, see Fig. 13(b). This evaporation leads to a notable pressure growth in the onboard tanks reaching 0.80 MPa ab in Tank 1, which is closer to the manifold, and a slightly lower pressure of 0.76 MPa ab in Tank 2, see Fig. 11.

The continuous sLH2 flow cools down the pipes. Hydrogen temperature approaches the recommended temperature corridor limits [7,8] at about 70 s. Then, hydrogen starts to enter the onboard tanks in a liquid state, see Fig. 13(c). The sLH2 cools the tanks and their content. The condensation rate inside the tanks increases and results in decreasing tank pressure, see Fig. 11.

Once the pipes and tanks are sufficiently cooled, step 2 of the main refuelling begins with an increasing mass flow rate up to 400 kg/h (111 g/s). During this step, the pressure inside the tanks rises slowly rather than sharply, see Fig. 11. This is due to the gradual compression of the gaseous hydrogen in the decreasing ullage space. Maintaining a controlled and gradual increase in pressure is crucial for the structural integrity of the tanks and the overall system's safety. The cooling effect of the sLH2 flow helps to prevent fast pressure increase in the pipes due to evaporation and ensures the tanks' stable operation.

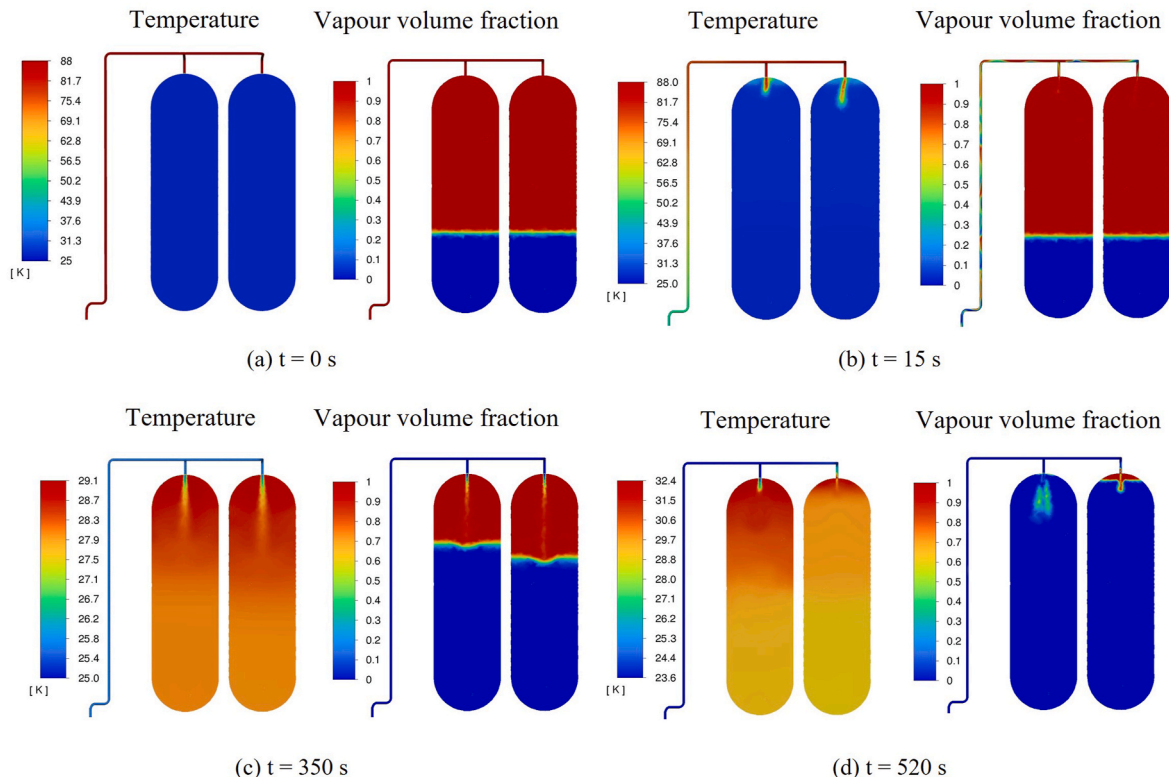


Fig. 13. The total temperature and vapour volume fraction contour plots in the cross-section of the tanks at various times.

When the tanks are nearly full, see Fig. 13(d), the pressure starts to increase sharply because LH2 has much lower compressibility than gaseous hydrogen. This sharp pressure rise indicates that the tanks have reached their maximum capacity. The automatic valve cuts off the flow when the tank pressure reaches the target of 1.6 ± 0.05 MPa [8].

The results of simulations align closely with the trends presented in the conceptual refuelling process (Fig. 2), which is based on experimental observations. This model serves as a valuable tool for developing efficient LH2 refuelling protocols, offering an affordable alternative to experimental testing, e.g. by pre-test simulations.

5. Conclusions

The originality of this work is in the development for the first time of the original modelling approach for LH2 refuelling through the entire equipment of HRS that includes a thermodynamic model for equipment from the HRS reservoir to the pump outlet, and a contemporary 3D two-phase CFD model from the pump outlet to the onboard storage tanks. The concept of efficient refuelling to SoC = 100% using sub-cooled liquid hydrogen is explained and verified through insights provided by the thermodynamic model and analysis of results of CFD simulations. The thermodynamic model is used to define the LH2 temperature released by the pump into the space of piping at pre-defined pressure which is used as an input to the CFD model. It was shown that the LH2 pump efficiency is an important parameter when designing LH2 refuelling station equipment and protocols which could directly affect the efficiency and overall safety of the HRS. The CFD model encompasses all the HRS components located downstream of the pump including bent pipes, PCV for mass flow rate control, breakaway, hose, nozzle, and two onboard storage tanks.

The significance of this study lies in the development of the model that can serve as a contemporary engineering tool for the design of efficient and inherently safer LH2 refuelling station equipment and refuelling protocols. The model demonstrates high flexibility, allowing for the modification or replacement of each system component to conduct parametric design studies. This study provides valuable insights into the underlying physical phenomena of heat and mass transfer during LH2 refuelling at HRS, enhancing our understanding and even visualisation of the process using CFD. The model can be used to design LH2 refuelling station components and optimise protocols.

The rigour of this work is in the verification of the CFD simulations against a conceptual LH2 refuelling process published elsewhere. Simulations were conducted using ANSYS Fluent as the computational engine, supplemented by an in-house developed UDF code to regulate flow rate through the PCV that was successfully used in our previous studies on gaseous hydrogen refuelling through the entire equipment of HRS. The PCV is simulated as a porous media with a sink term in the momentum conservation equations, which was controlled by varying the media porosity. The successful verification of the model against the conceptual refuelling protocol confirms its accuracy and reliability, making it a robust tool for future research and development of LH2 transfer technologies. Further, the validation of the model will be conducted as experimental data on LH2 refuelling becomes available.

CRedit authorship contribution statement

Vladimir Molkov: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Hazhir Ebne-Abbasi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Dmitriy Makarov:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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